

# **STUDENT PERFORMANCE ANALYTICS & DECISION SUPPORT SYSTEM**

## **1. Purpose**

This document defines the functional behavior expected from the analytics solution.

## **2. Functional Requirements**

### **Module 1: Student Management**

#### **FR001: Student Data Management**

##### **Description:**

The system shall maintain student master information.

##### **Inputs:**

- Student ID
- Name
- Class
- Section
- Subject details

Expected Output: Centralized student database.

#### **FR002: Student Identification**

The system shall ensure every student has a unique identifier.

##### **Acceptance:**

- Duplicate student IDs should not exist.

### **Module 2: Test Management**

#### **FR003: Test Information Management**

The system shall store:

- Test ID
- Test name
- Subject
- Date
- Difficulty level
- Total marks

### **Module 3: Question Management**

#### **FR004: Question Bank Management**

The system shall maintain:

- Question ID
- Subject
- Chapter
- Difficulty
- Marks

### **Module 4: Response Management**

#### **FR005: Student Response Tracking**

The system shall capture:

- Student ID
- Test ID
- Question ID
- Response status

### **Module 5: Score Calculation**

#### **FR006: Automated Score Calculation**

The system shall calculate:

- Correct answers
- Incorrect answers
- Percentage
- Grade

#### **Formula:**

Percentage = (Correct Answers / Total Questions) × 100

### **Module 6: Performance Analysis**

#### **FR007: Student Progress Tracking**

The system shall provide:

- Average score
- Best score
- Lowest score
- Improvement trend

### **FR008: Risk Identification**

The system shall classify students based on performance.

#### **Categories:**

| Category       |
|----------------|
| Top Performer  |
| High Performer |
| Average        |
| Needs Support  |
| At Risk        |

### **Module 7: Analytics**

#### **FR009: Question Analytics**

The system shall analyze:

- Question difficulty
- Accuracy rate
- Student success rate

#### **FR010: Chapter Analytics**

The system shall analyze:

- Chapter performance
- Subject performance
- Learning gaps

### **Module 8: Dashboard**

#### **FR011: Executive Dashboard**

The system shall display:

- Total students
- Average score
- Pass percentage
- Risk students

#### **FR012: Teacher Dashboard**

The system shall display:

- Subject performance
- Chapter performance
- Weak students